Discrete Mathematics (Y1S1) August 23, 2026

Discrete Mathematics: Concept Checklist

Active-Recall & Self-Testing Revision Guide

1 Logic, Quantifiers, and Proof Methods

[]**Conventions & Exam Pitfalls.**

- ▫ **Linguistic Conjunction:** In mathematical logic, the word ***"but"*** functions as conjunction ($\wedge$). Example: "p is true but q is false" $\equiv p \wedge \neg q$.
- ▫ **Inference Declarations:** When doing formal inference, always explicitly state what propositions each variable p, q, r represents.
- ▫ **Universal Proofs:** When proving $\forall x \in S$, begin with "***Given an arbitrary*** $x \in S$" or "***Let*** $x \in S$ ***be arbitrary***".

[]**Definitions & Concepts to State 1.1**

- ▫ **Exclusive OR ($\oplus$):** Write down the full truth table of $p \oplus q$. Express $p \oplus q$ in terms of $\wedge, \vee, \neg$.
- ▫ **Conditional Statement ($p \implies q$):** List at least 8 equivalent English phrasings of $p \implies q$, including:
  - "q follows from p"
  - "q unless $\neg p$"
  - "p only if q", "p is sufficient for q", "q is necessary for p", etc.
- ▫ **Variants of Conditional:** State the Converse ($q \implies p$), Contrapositive ($\neg q \implies \neg p$), and Inverse ($\neg p \implies \neg q$). Which are logically equivalent?
- ▫ **Quantifiers with Restricted Domains:** State the logical expansion for:

$$\forall x < 0\, P(x) \quad \text{and} \quad \exists x > 0\, Q(x).$$

- ▫ **Nested Quantifiers:** Explain why the order of quantifiers matters. Compare $\forall x \exists y\, P(x, y)$ vs. $\exists y \forall x\, P(x, y)$ with an example. (Review Assignment 1 nested quantifier problems).
- ▫ **Special Proof Methods:**
  - What is a ***vacuous proof***? When does it apply?
  - What is a ***trivial proof***?
  - What is an ***exhaustive proof*** (proof by cases)?

[]**Theorems & Identities to State 1.2**

- □ **Core Logic Laws:** State De Morgan's laws, Distributive laws, Absorption laws, and Conditional equivalence.
- □ **8 Rules of Inference for Propositional Logic:** List all 8 rules and write their tautological forms:
  - Modus Ponens, Modus Tollens, Hypothetical Syllogism, Disjunctive Syllogism
  - Addition, Simplification, Conjunction, Resolution
- □ **4 Rules of Inference for Quantified Statements:** State Universal Instantiation (UI), Universal Generalization (UG), Existential Instantiation (EI), and Existential Generalization (EG).

[]**Proof Challenges & Calculations 1.3**

- □ **Quantifier Distribution (Proofs):**
  1. Prove: $\forall x\,(P(x) \wedge Q(x)) \equiv \forall x\,P(x) \wedge \forall x\,Q(x)$.
  2. Prove: $\exists x\,(P(x) \vee Q(x)) \equiv \exists x\,P(x) \vee \exists x\,Q(x)$.
- □ **Quantifier Distribution (Disproofs & Counterexamples):**
  1. Disprove: $\forall x\,(P(x) \vee Q(x)) \equiv \forall x\,P(x) \vee \forall x\,Q(x)$.
  2. Disprove: $\exists x\,(P(x) \wedge Q(x)) \equiv \exists x\,P(x) \wedge \exists x\,Q(x)$.
- □ **Set Identity Proof via Logic:** Let $A, B \subseteq U$. Prove that $A \subseteq B \Leftrightarrow \overline{A} \cup B = U$ by translating to propositional equivalences.

2 Sets, Functions, and Cardinality

[]**Conventions & Exam Pitfalls.**

- □ **Natural Numbers Convention:** In this course, $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$. $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$.
- □ **Set $\leftrightarrow$ Logic Translation:** $U \leftrightarrow \mathbf{T}$ (True), $\varnothing \leftrightarrow \mathbf{F}$ (False), $\cup \leftrightarrow \vee$, $\cap \leftrightarrow \wedge$, $\overline{A} \leftrightarrow \neg A$.

[]**Definitions & Concepts to State 2.1**

- ▫ **Set Terminology:** Define singleton set, proper subset ($A \subsetneq B$), power set $\mathscr{P}(A)$, and Cartesian product $A \times B$.
- ▫ **Set Operations:** State formal set-builder definitions of $A \cup B$, $A \cap B$, $A \backslash B$, and $\overline{A}$.
- ▫ **Function Classifications:** Define real-valued function vs. integer-valued function.
- ▫ **Image & Preimage:** For $f : A \to B$, define $f(S)$ for $S \subseteq A$, and $f^{-1}(T)$ for $T \subseteq B$.
- ▫ **Injectivity, Surjectivity, Bijectivity:** Write the formal logical definitions of:
  - Injective (one-to-one)
  - Surjective (onto)
  - Bijective (one-to-one correspondence)
  - Invertible function and inverse function f^{-1}
- ▫ **Floor & Ceiling Characterizations:** State the fundamental bounding inequalities:

$$\lfloor x \rfloor \leqslant x < \lfloor x \rfloor + 1 \quad \text{and} \quad \lceil x \rceil - 1 < x \leqslant \lceil x \rceil.$$

- ▫ **Rounding Function:** Express round(x) in terms of $\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$.
- ▫ **Countability Characterization:** State the three equivalent characterizations for a set $S \neq \varnothing$ to be countable.

[]**Theorems & Identities to State 2.2**

- ▫ **Function Composition Properties ($f : A \to B, g : B \to C$):** State the 4 major composition theorems regarding injectivity and surjectivity of $g \circ f$.
- ▫ **Invertibility Theorem:** State the condition for f to be invertible (f invertible $\Leftrightarrow$ f bijective).
- ▫ **Injection-Surjection Duality:** Show that for non-empty sets, ($\exists$ injection $f : A \to B$) $\Leftrightarrow$ ($\exists$ surjection $g : B \to A$).
- ▫ **Countable Operations:** State the closure properties of countable sets under subsets, finite unions, countable unions, and Cartesian products.
- ▫ **Cantor's Theorem:** State Cantor's theorem comparing $|A|$ and $|\mathscr{P}(A)|$.

[]**Proof Challenges & Calculations 2.3**

- ▫ **Composition Inverses Counterexamples:**
  1. Give an example where $g \circ f$ is injective, but g is ***not*** injective.
  2. Give an example where $g \circ f$ is surjective, but f is ***not*** surjective.

- ▫ **Countability Proofs:**
  - Prove $\mathbb{Z}^+ \times \mathbb{Z}^+$ is countably infinite.
  - Prove $\mathbb{Q}$ is countably infinite.
  - Prove $\mathbb{R}$ is uncountable via Cantor's diagonal argument.
  - Prove $|A| < |\mathscr{P}(A)|$ for all sets A.

- ▫ **Function Space Bijection:** Let $S \neq \varnothing$ and $F_S = \{f : S \to \{1, 2\}\}$. Prove there is a bijection between F_S and $\mathscr{P}(S)$.

- ▫ **Cardinality Computations ($|A| = m$):** Evaluate:
  - $|\{x \in \mathscr{P}(A) : |x| \leqslant 1\}|$
  - $|\{x \subseteq \mathscr{P}(A) : |x| \leqslant 1\}|$

- ▫ **Equinumerosity Bijections:** Construct explicit bijections showing:
  - $|\mathbb{R}| = |(0, \infty)|$
  - $|\mathbb{R}| = |(\sqrt{2}, \infty)|$
  - $|\mathbb{R}| = |(0, 1)|$
  - $|[0, 1]| = |(0, 1)|$
  - $|\mathscr{P}(\mathbb{Z})| = |\mathscr{P}(\mathbb{Z}^+)|$

- ▫ **Subsets & Partitions of Reals / Naturals:**
  - Prove or disprove: There exists a countably infinite subset of the set of irrational numbers.
  - Describe a partition of $\mathbb{N}$ that divides $\mathbb{N}$ into $\aleph_0$ countably infinite subsets.

[]**Textbook Exercise Checklist.**

Section	**Textbook Problems**	**Status**
Section 2.1	Problem 27	□ Done
Section 2.2	Problem 17	□ Done
Section 2.3	Problems 35, 45	□ Done
Section 2.5	Problem 11	□ Done

3 Algorithms and Asymptotic Analysis

[]**Conventions & Exam Pitfalls.**

- □ **Logarithm Notation:** In this course, $\log x = \ln x$ unless specified otherwise.
- □ **Disproving Big-O Strategy:** To prove $f(x) \neq O(g(x))$, use proof by contradiction: assume witnesses C, k exist, and derive that an unbounded function is bounded.

[]**Definitions & Concepts to State 3.1**

- □ **Binary Search:** Explain the binary search mechanism and state its recurrence and asymptotic complexity.
- □ **Asymptotic Definitions:** State the formal definition of:
  - Big-O notation: $f(x) = O(g(x))$ with witnesses C, k.
  - Big-Ω notation: $f(x) = \Omega(g(x))$.
  - Big-Θ notation: $f(x) = \Theta(g(x))$.
  - Little-o notation: $f(x) = o(g(x))$.

[]**Theorems & Identities to State 3.2**

- ▫ **Limit Criteria:** State the limit test for O, Ω, Θ, o when $L = \lim_{x\to\infty} \left|\frac{f(x)}{g(x)}\right|$.
- ▫ **Sum & Product Rules:** State the sum rule $(f_1 + f_2)(x) = O(\max(|g_1|, |g_2|))$ and product rule $(f_1 f_2)(x) = O(g_1 g_2)$.
- ▫ **Symmetry of Θ:** State why $f(x) = \Theta(g(x)) \Leftrightarrow g(x) = \Theta(f(x))$.
- ▫ **Duality of O and Ω:** State why $f(x) = O(g(x)) \Leftrightarrow g(x) = \Omega(f(x))$.

[]**Proof Challenges & Calculations 3.3**

- ▫ **Witness Finding:** Prove $f(x) = x^2+2x+1$ is $O(x^2)$ by finding explicit constants C and k.
- ▫ **Disproving Big-O:** Prove by contradiction that n^2 is not $O(n)$.
- ▫ **Growth Comparison Tasks:**
  - Prove that for any $n \in \mathbb{N}$ and $a > 0$, the order of x^n is smaller than e^{ax} ($x^n = o(e^{ax})$).
  - Compare the asymptotic orders of x^4 and $x^3 \log^{100} x$.
  - Prove that $\log(n!) = \Theta(n \log n)$.

[]**Textbook Exercise Checklist.**

Section	Textbook Problems	Status
Section 3.2	Problems 3, 4, 9, 20, 47, 67, 71, 73	▫ Done

4 Number Theory and Cryptography

[]**Conventions & Exam Pitfalls.**

- ▫ **Modular Inverses Requirement:** a has an inverse modulo $m \Leftrightarrow \gcd(a, m) = 1$.
- ▫ **Euler's Totient Requirement:** $a^{\varphi(n)} \equiv 1 \pmod{n}$ holds if and only if $\gcd(a, n) = 1$.

[]**Definitions & Concepts to State 4.1**

- ▫ **Divisibility:** Define $a \mid b$. Define quotient and remainder in the Division Algorithm.
- ▫ **Congruence Modulo m:** Define $a \equiv b \pmod{m}$ and congruence class $[a]_m$.
- ▫ **Prime Concepts:** Define prime, composite, Mersenne prime, relatively prime, and pairwise relatively prime.
- ▫ **Sieve of Eratosthenes:** Briefly explain how the Sieve of Eratosthenes identifies all primes up to N.
- ▫ **GCD and LCM:** Write the definitions and characterizations of $\gcd(a, b)$ and $\operatorname{lcm}(a, b)$.
- ▫ **Bézout's Theorem:** State Bézout's identity for $\gcd(a, b)$.
- ▫ **Euler's Totient Function:** Define $\varphi(n)$. State the formula for $\varphi(n)$ and $\varphi(pq)$ for distinct primes.

[]**Theorems & Identities to State 4.2**

- ▫ **Divisibility Properties:** State linear combination divisibility: $a \mid b \wedge a \mid c \implies a \mid (mb + nc)$.
- ▫ **Division Algorithm:** State the formal existence and uniqueness theorem for $a = dq + r$.
- ▫ **Modular Arithmetic Properties:** State addition, multiplication, and cancellation rules modulo m.
- ▫ **Prime Factor Bound:** State the theorem: if n is composite, it has a prime factor $p \leqslant \sqrt{n}$.
- ▫ **Euclidean Reduction:** State $\gcd(a, b) = \gcd(b, r)$ where $a = bq + r$.
- ▫ **Euclid's Lemma:** If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.
- ▫ **Fundamental Theorem of Arithmetic:** State the unique prime factorization theorem.
- ▫ **GCD-LCM Product Theorem:** State $\gcd(a, b) \cdot \mathrm{lcm}(a, b) = |ab|$.
- ▫ **Chinese Remainder Theorem (CRT):** State the general CRT for pairwise coprime moduli $m_1, \ldots, m_n$.
- ▫ **Fermat's Little Theorem & Euler's Theorem:** State both theorems.
- ▫ **RSA Cryptosystem Principle:** State the RSA identity $m^{de} \equiv m \pmod{n}$ for $n = pq$ and $de \equiv 1 \pmod{\varphi(n)}$.

[]**Proof Challenges & Calculations 4.3**

- **Number Theory Proofs:**
  - Prove: If $a, b \neq 0$ and $a \mid b$, then $|a| \leqslant |b|$.
  - Prove: $a \equiv b \pmod{m} \Leftrightarrow (a \bmod m) = (b \bmod m)$.
  - Prove that every integer > 1 has a prime factor.
  - Prove: If $2^m - 1$ is prime, then m is prime.
  - Prove Euclid's Theorem: There are infinitely many primes.
  - Prove that if p is prime, then either $p \mid a$ or $\gcd(a, p) = 1$.
  - Prove: If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.
  - Prove: $\gcd(k, mn) = 1 \Leftrightarrow \gcd(k, m) = 1 \wedge \gcd(k, n) = 1$.
  - Prove the Chinese Remainder Theorem.
  - Prove $\varphi(pq) = (p-1)(q-1)$ for distinct primes p, q.
  - Prove: $\lceil n/k \rceil = \lfloor (n-1)/k \rfloor + 1$ for $n, k \in \mathbb{Z}^+$.
- **Calculations & Algorithms:**
  - Use the Euclidean Algorithm to compute $\gcd(1236, 2136)$.
  - Find the modular inverse of a number using the Extended Euclidean Algorithm.
  - Solve the system via CRT: $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$.
  - Find all solutions to the quadratic congruence $x^2 \equiv 29 \pmod{35}$.

[]**Textbook Exercise Checklist.**

Section	**Textbook Problems**	**Status**
Section 4.1	Problems 19, 21, 22, 23	☐ Done
Section 4.3	Problems 32, 43	☐ Done
Section 4.4	Problems 11, 21, 40, 65	☐ Done

5 Induction and Recurrence Relations

[]**Definitions & Concepts to State 5.1**

- **Induction Templates:** Outline the standard steps of:
  - Principle of Mathematical Induction (Weak Induction).
  - Principle of Strong Mathematical Induction.
- **Well-Ordering Principle:** State the Well-Ordering Principle for non-negative integers.
- **Linear Recurrences Definitions:** Define Linear Homogeneous Recurrence Relations with Constant Coefficients (LHRRWCC) and their characteristic equations.

[]**Theorems & Identities to State 5.2**

- **Distinct Roots Theorem:** State the general solution form for $a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}$ when characteristic roots $r_1, \ldots, r_k$ are distinct.
- **Repeated Roots Theorem:** State the solution form when root r_i has multiplicity m_i.
- **Linearity of Recurrence Solutions:** State the superposition principle for homogeneous recurrences.
- **Nonhomogeneous Solution Structure:** State why the general solution is $a_n = a_n^{(p)} + a_n^{(h)}$.

[]**Proof Challenges & Calculations 5.3**

- □ **Prime Factorization via Strong Induction:** Write the full inductive proof that every integer $n > 1$ can be factored into primes.
- □ **Superposition Proofs:**
  - Prove that if $\left\{a_n^{(1)}\right\}, \left\{a_n^{(2)}\right\}$ solve the homogeneous recurrence, then $\alpha a_n^{(1)} + \beta a_n^{(2)}$ is also a solution.
  - Prove that if $\left\{a_n^{(1)}\right\}, \left\{a_n^{(2)}\right\}$ solve the nonhomogeneous recurrence, then $\left\{a_n^{(1)} - a_n^{(2)}\right\}$ solves the homogeneous equation.
- □ **Characteristic Root Equivalence:** Prove that $a_n = r^n$ $(r \neq 0)$ is a solution to $a_n = \sum_{i=1}^{k} c_i a_{n-i} \iff r^k - \sum_{i=1}^{k} c_i r^{k-i} = 0$.
- □ **Recurrence Calculations:**
  - Solve $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with $a_0 = 2, a_1 = 5, a_2 = 15$.
  - Solve $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$ with $a_0 = 1, a_1 = 4, a_2 = 28$.

[]**Textbook Exercise Checklist.**

Section	**Textbook Problems**	**Status**
Section 5.1	Problems 19, 21, 22	□ Done

6 Counting and Combinatorics

[]**Definitions & Concepts to State 6.1**

- □ **Basic Counting Principles:** State formulas for total functions $A \to B$ (n^m) and one-to-one functions $P(n, m)$.
- □ **Pigeonhole Principle:** State the basic and Generalized Pigeonhole Principle.
- □ **Permutations & Combinations:** State formulas for $P(n, r)$, $\binom{n}{r}$, and multinomial coefficients $\binom{n}{n_1, \ldots, n_k}$.
- □ **Derangements:** Define derangement (D_n) and state its summation formula.

[]**Theorems & Identities to State 6.2**

- ▫ **Binomial Theorem:** State $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k$.
- ▫ **Combinatorial Identities to Know:**
  - Pascal's Identity: $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$
  - Sum of Coefficients: $\sum_{k=0}^{n} \binom{n}{k} = 2^n$
  - Vandermonde's Identity: $\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k} \binom{n}{k}$
  - Sum of Squares: $\sum_{k=0}^{n} \binom{n}{k}^2 = \binom{2n}{n}$
  - Hockey-Stick Identity: $\sum_{k=r}^{n} \binom{k}{r} = \binom{n+1}{r+1}$
- ▫ **Principle of Inclusion-Exclusion (PIE):** State the general formula for $|\bigcup_{i=1}^{n} A_i|$.
- ▫ **Surjective Functions Count:** State formula for onto functions from size m to size n: $\sum_{j=0}^{n} (-1)^j \binom{n}{j} (n-j)^m$.

[]**Proof Challenges & Calculations 6.3**

- ▫ **Combinatorial Proofs:**
  - Provide an algebraic and a combinatorial proof for Pascal's Identity.
  - Give a combinatorial proof for the Hockey-Stick Identity.
  - Prove $\varphi(mn) = \varphi(m)\varphi(n)$ for $\gcd(m, n) = 1$ using PIE.
- ▫ **Stars and Bars Problems:**
  - Find non-negative integer triples (x_1, x_2, x_3) satisfying $x_1 + x_2 + x_3 = 10$.
  - Find positive integer triples $(x_i \geqslant 1)$ satisfying $x_1 + x_2 + x_3 = 10$.
- ▫ **Combinatorial Problems:**
  - Calculate onto functions from a set of 6 elements to a set of 3 elements.
  - Calculate ways 5 friends in Secret Santa can draw names such that nobody draws their own name (D_5).
  - How many strictly alphabetical length-5 lists can be made from $\{A, B, C, D, E, F, G, H, I\}$?
  - How many permutations of $\{A, B, C, D, E, F\}$ have D occurring before A?
  - Prove that for any $a \in \mathbb{N}$, there exist distinct $k, l \in \mathbb{N}$ such that $10 \mid (a^k - a^l)$ via Pigeonhole Principle.

7 Relations, Equivalence Relations, and Posets

[]**Definitions & Concepts to State 7.1**

- ▫ **Relation Definitions:** Define relation R on set A. How many relations exist on an n-element set?
- ▫ **Relation Properties:** Write formal definitions of:
  - Reflexive
  - Symmetric
  - Antisymmetric
  - Transitive
- ▫ **Representations:** Describe zero-one matrix representation M_R and digraphs.
- ▫ **Composition & Transitive Closure:** Define $S \circ R$, Boolean product $M_R \odot M_S$, and connectivity relation R^*.
- ▫ **Equivalence Relations:** Define equivalence relation and equivalence classes $[a]$.
- ▫ **Poset Fundamentals:** Define partial ordering ($\preceq$), comparability, total order (chain), and Hasse diagram.
- ▫ **Extremal Elements:** Define maximal, minimal, greatest, least, upper bound, lower bound, LUB (supremum), and GLB (infimum).

[]**Theorems & Identities to State 7.2**

- ▫ **Transitivity Criterion:** State R is transitive $\Leftrightarrow$ $R^n \subseteq R$ for all $n \geqslant 1$ $\Leftrightarrow$ $R^2 \subseteq R$.
- ▫ **Fundamental Theorem of Equivalence Relations:** State the partition-equivalence duality theorem.
- ▫ **Equal-Class Cardinality Theorem:** State the relation $|R| = m|A|$ when all classes have size m.
- ▫ **Composition Monotonicity:** State $R_1 \subseteq R_2 \implies S \circ R_1 \subseteq S \circ R_2$.

[]**Proof Challenges & Calculations 7.3**

- ▫ **Properties Verification:** Determine whether $R = \{(a,b) \in \mathbb{Z} \times \mathbb{Z} : a = b + 1\}$ is antisymmetric.
- ▫ **Matrix Composition Proof:** Prove $M_{S \circ R} = M_R \odot M_S = [t_{ij}]$ where $t_{ij} = \bigvee_k (r_{ik} \wedge s_{kj})$.
- ▫ **Equal-Class Proof:** Prove that if every equivalence class of R on A has size m, then $|R| = m|A|$.
- ▫ **Poset Analysis:** In $(\mathscr{P}(\{1,2,3\}), \subseteq)$:
  - Identify greatest and least elements.
  - Find upper bounds, lower bounds, LUB, and GLB for $A = \{\{1\}, \{2\}\}$.